(called “three-branes”), and a host of other ingredients as well. As with the eleventh dimension, this feature of M-theory emerges when calculations are freed from reliance on the approximations used prior to the mid-1990s.

Beyond these and a variety of other insights attained over the last few years, much of the true nature of M-theory remains mysterious–one suggested meaning for the “M.” Physicists worldwide are working with great vigor to acquire a full understanding of M-theory, and this may well constitute the central problem of twenty-first-century physics.

An Approximation Method

The limitations of the methods physicists have been using to analyze string theory are bound up with something called perturbation theory. Perturbation theory is an elaborate name for making an approximation to try to give a rough answer to a question, and then systematically improving this approximation by paying closer attention to fine details initially ignored. It plays an important part in many areas of scientific research, has been an essential element in understanding string theory, and, as we now illustrate, is also something we encounter frequently in our day-to-day lives.

Imagine that one day your car is acting up, so you go see a mechanic to have it checked out. After giving your car a once-over, he gives you the bad news. The car needs a new engine block, for which parts and labor typically run in the $900 range. This is a ballpark approximation that you expect to be refined as the finer details of the work required become apparent. A few days later, having had time to run additional tests on the car, the mechanic gives you a more precise estimate, $950. He explains that you also need a new regulator, which with parts and labor costs about $50. Finally, when you go to pick up the car, he has added together all of the detailed contributions and presents you with a bill of $987.93. This, he explains, includes the $950 for the engine block and regulator, an additional $27 covering a fan belt, $10 for a battery cable, and $.93 for an insulated bolt. The initial approximate figure of $900 has been refined by including more and more details. In physics terms, these details are referred to as perturbations to the initial estimate.